Category:

Crypto

Name:

C5: A very old odd file

Message:

Description:

Earlier communication and data transmission utilizes the telegraph system. It uses electrical signals over wires. The sender uses a telegraph key to interrupt the electrical current, which was then decoded at the receiving end. There are various encoding systems used in telegraph systems and the most famous is the Morse code.

Details:

Old computer veteran from the days of cold war left nothing but a text file. It puzzles us as it contains just one sentence and some random characters. Will you help us to try to find his secret?

Since we're so nice, we'll even give you a free hint! The flag value will
GSC_FLAG{LOOK.SOMETHING.LIKE.THIS}

Instructions:

To extract the data you need to convert
0x09 and 0x0A into '0' and '1'

In UNIX we do it for example like this:

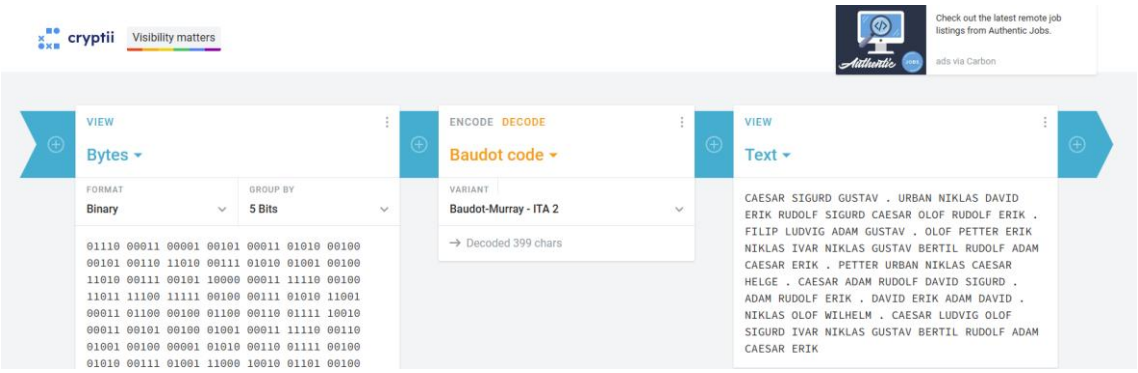
```
sed   's/-/0/g; s/\./1/g'   < C6_a_very_old_odd_file.txt > output.txt
```

Now we have a ton of zeros and ones. It is not ASCII, nor HEX. The old veteran used a lot older encoding.
First, we split the binary not to 8bit nor 7bit words, but 5bit words (Does it ring a bell yet?)

Now what is 5bit encoding used for? Ah, good old (remeber the veterans) punch tape!
Baudot encoding is rather simple an so you can:
a) solve it by hand. (Naah, too lazy)
b) code it from the beginning (hmmm, lazy too)

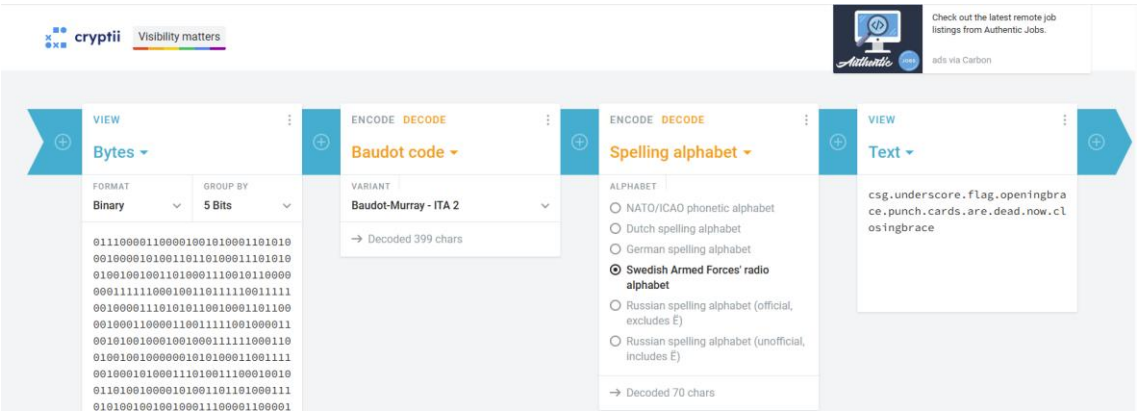
c) use Python library / https://python-baudot.readthedocs.io/en/latest/index.html (yeah, CTF ninja)

d) use https://cryptii.com/



WTF? Strange, right? Well not all spelling alphabets are NATO. There are some other like Swedish for example.

Let’s run the text through the Cryptii.com again, this time with Swedish spelling.



You should get “csg.underscore.flag.openingbrace.punch.cards.are.dead.new.closingbrace”.

Hm, what is "opening / closing brace" ? Well Baudot encoding does not contain curly braces '{}'.

And we know, that the flags are in CSG_FLAG{xxx} format. So the solution is:

CSG_FLAG{PUNCH.CARDS.ARE.DEAD.NOW}